

Linear systems and elimination

Forsythe's conjecture beyond restart length two

Difficulty: extreme **Importance:** interesting to the community

Status: open; the case of restart length two has a recent proof claim.

Problem statement

Let $A \in \mathbb{R}^{n \times n}$ be symmetric positive definite, $b, x_0 \in \mathbb{R}^n$, and $3 \leq s < n$. At each restart, perform exactly s exact-arithmetic conjugate-gradient steps, starting from the last iterate, and discard the previous search directions. Denote the iterate after restart cycle j by x_j . Suppose this process never terminates exactly, and put

$$r_j = b - Ax_j, \quad y_j = r_j / \|r_j\|_2.$$

Prove or disprove that both $(y_{2j})_{j \geq 0}$ and $(y_{2j+1})_{j \geq 0}$ converge in $\mathbb{R}^n$, for every such input and restart length. The question concerns directions, not whether the residual norms tend to zero. Exact termination is excluded so every normalization exists.

References

Faber, Liesen, and Tichý, *On the Forsythe conjecture*, BIT 63 (2023), §2, especially the displayed Forsythe conjecture. Colbrook, Stepaniants, and Townsend, *A Proof of the Forsythe Conjecture for the Two-Step Restarted Conjugate Gradient Method*, August 2026, §1 and main theorem.

Status check

Searches for *Forsythe conjecture 2026 proof* found the August preprint. Its claimed theorem covers $s = 2$, not all $s \geq 3$. The surviving range is therefore stated explicitly; older claims that every $s \geq 2$ remains open are obsolete.